SplineSketch: Even More Accurate Quantiles with Error Guarantees

Aleksander Łukasiewicz[1], Jakub Tětek, Pavel Veselý[1]

[1]Computer Science Institute of Charles University, Prague, Czechia

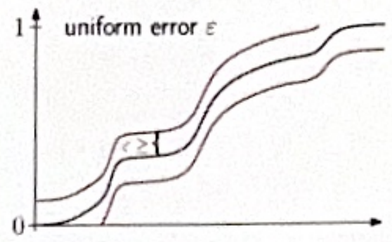
Compared to *t*-digest

1. Introduction & Motivation

The challenge: characterize the distribution of numerical data with low memory footprint & high accuracy

Quantile summary: data structure of sublinear size

- processes n items in one pass – streaming algo
- approximates ranks & quantiles with error $\pm\varepsilon \cdot n$
  - $\text{rank}(y) = \#$ input items $\leq y$
  - for $\phi \in [0,1]$, a ϕ-quantile is the $\lceil \phi \cdot n \rceil$-smallest item
- mergeable $\Rightarrow$ allow distributed or parallel processing



Current State of Practice: t-digest [Dunning Softw Impacts'21]: Widely used in open source & tech companies. Offers high accuracy in practice but lacks error guarantees & worst-case error can be arbitrarily bad [Cormode et al. KDD'21].

Theoretical Approaches: KLL [Karnin-Lang-Liberty FOCS'16] or GK [Greenwald-Khanna SIGMOD'01] sketches offer uniform $\pm\varepsilon n$ error bounds and use asymptotically optimal space but do not match practical accuracy of t-digest.

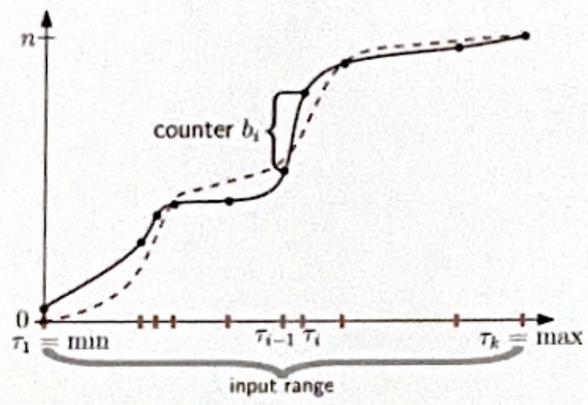
2. SplineSketch: The Best of Both Worlds

Objective: Provide uniform error guarantees, fixed memory consumption (single parameter k), mergeability, and outperform t-digest in practice.

Two Main Ingredients:

1. **Dynamic Subdivision into k Buckets:** Maintained online via thresholds $\tau_1, \ldots, \tau_k$ and counters $b_1, \ldots, b_k$.
2. **Cubic Spline Interpolation over Buckets:** Uses piecewise cubic Hermite interpolating polynomial (**PCHIP**) to reconstruct the CDF

Invariant: always exactly k buckets.



3. Maintaining Buckets: Splits & Joins

Goal: maintain buckets s.t. the interpolation captures the CDF well

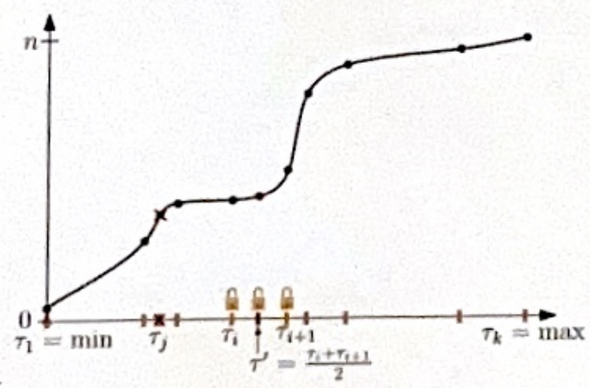
- and worst-case error bounded

Split: add new threshold in the middle

- counter b_i split based on interpolation

Join adjacent buckets

- remove the threshold in between
- new counter = sum of the two counters



Other ingredients:

- **Bucket bound:** counters $b_i \leq C_b \cdot \frac{n}{k}$ for a suitable const. C_b
- **Heuristic error** to decide joins & splits, based on a discrete version of **2nd derivative of the CDF**, namely $\max\left(\frac{\left|\frac{b_i}{\ell_i} - \frac{b_{i-1}}{\ell_{i-1}}\right|}{\ell_i + \ell_{i-1}}, \frac{\left|\frac{b_{i+1}}{\ell_{i+1}} - \frac{b_i}{\ell_i}\right|}{\ell_{i+1} + \ell_i}\right) \cdot \ell_i^2$ where $\ell_i := \tau_{i+1} - \tau_i$ is bucket length
- **Protected thresholds** 🔒 to avoid repeated splits & joins
- **Epochs** of exp. increasing lengths – removing protection at the end of epoch

Joining Criteria: choose two adjacent buckets minimizing the heuristic error after join

- Only done if the resulting counter $< 0.75 \cdot C_b \cdot \frac{n}{k}$
- No join over a protected threshold

Splitting Criteria: either bucket bound exceeded: $b_i > C_b \cdot \frac{n}{k}$,

- or if the bucket has high heuristic error but only if:
  - $\geq k/3 + 2$ joinable pairs, b_i not too small, and
  - the heuristic error of the bucket is ≥ 1.5 times larger than the heuristic error of best joinable pair

4. Theoretical Guarantees

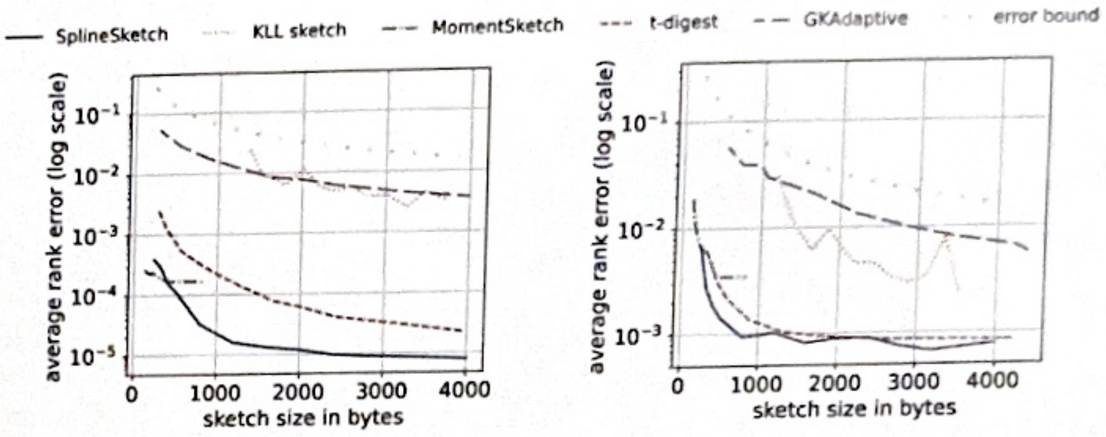
SplineSketch provides deterministic worst-case error bounds, in contrast to practical sketches such as t-digest or MomentSketch [Gan et al. VLDB'18].

- **Error Bound:** $O(\log(\alpha) \cdot \frac{n}{k})$, where α is the aspect ratio of the data
  - the analysis assumes filtering of items with frequency $> n/k$, can be done e.g. by Misra-Gries algo.
- **Aspect Ratio:** $\alpha = \frac{\text{max} - \text{min}}{\text{min diff. of distinct items}}$
- **Mergeability:** The analysis extends to merging sketches of a similar size.
- **Amortized Update Time:** $O(\log k)$ using a buffer of size k & heaps

5. Experimental Evaluation

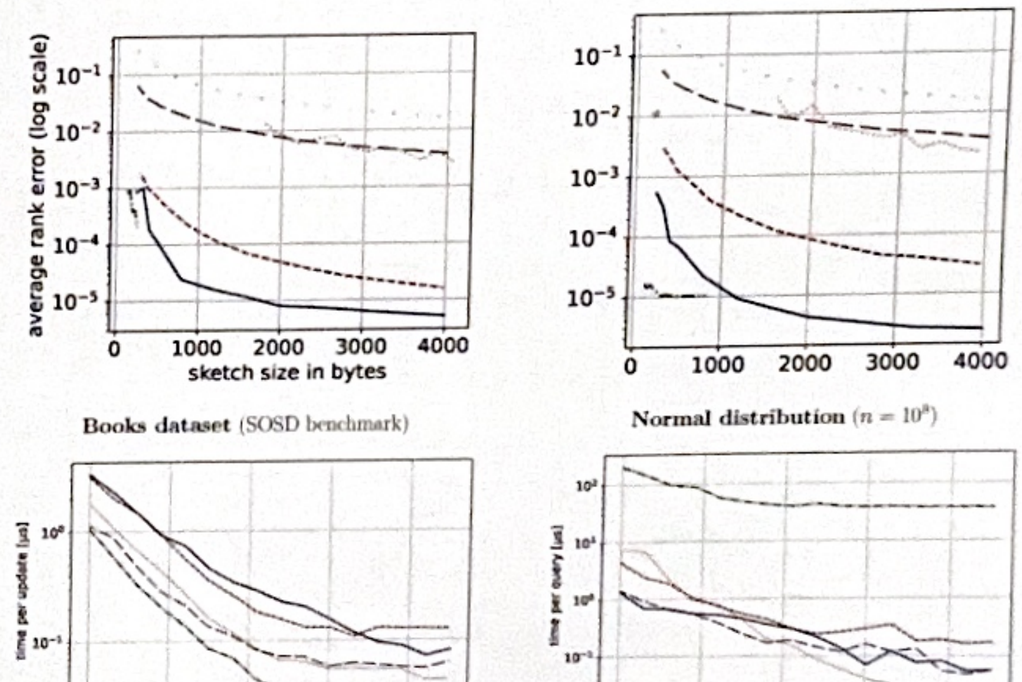
Evaluated SplineSketch and SOTA sketches on a range of real-world and synthetic datasets. Sample results for average rank error depending on the sketch size:

(error averaged over 10^5 evenly spaced queries)



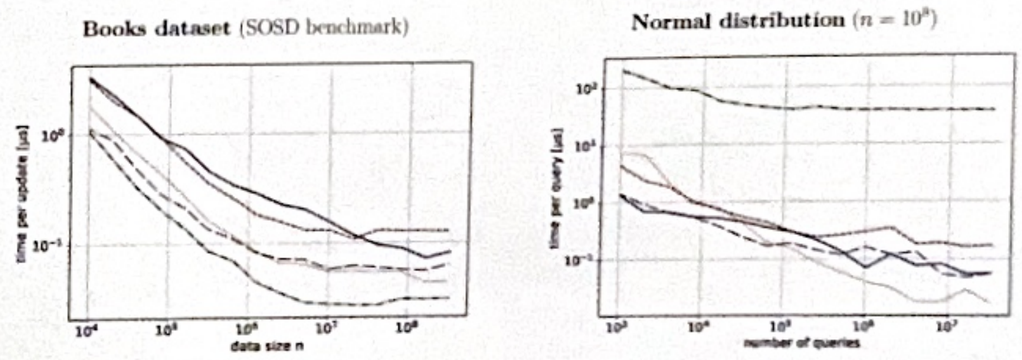
HEPMASS dataset (UCI ML repo., $n = 10,500,000$)

Power dataset (UCI ML repo., $n \approx 2$M)



Books dataset (SOSD benchmark)

Normal distribution ($n = 10^8$)



Update time (normal distribution, fixed sketch size of ≈ 1.6 kB)

Query time (normal distribution, fixed sketch size of ≈ 1.6 kB)

Bottom line: error typically much better than n/k

- typically 2–20 times better than t-digest

6. Summary: Pros and Cons

Advantages of SplineSketch:

- **High accuracy** across a range of synthetic and real-world datasets
- **Resizability:** can increase or decrease space usage at any time
- **Flexibility:** e.g., prior knowledge of distribution $\rightarrow$ even higher accuracy

Issues with SplineSketch:

- **Frequent Items:** Buckets do not work well if frequent items added; can be solved by heavy-hitter filtering (e.g., using Misra-Gries algo.)
- Numerical inputs with many significant digits (e.g. SOSD benchmark)
- **Worst-case error under full mergeability:** Currently, we can provide error bounds in streaming or if sketches of similar size are merged; an additional $\log(n/k)$ factor seems necessary under arbitrary merging.
- **Speed:** Larger update time than KLL or MomentSketch but similar to t-digest

7. Open Problems & Future Work



- **Inclusion in Apache DataSketches** library
- **Theoretical Gap:** 1. Observed empirical error is much better than the worst-case theoretical prediction. 2. Can we improve $\log(\alpha)$ to $\log\log(\alpha)$ in the error bound?
- **Relative Error:** Extending SplineSketch to provide relative-error guarantees, i.e., higher accuracy for the tails
- **User-defined Guarantees:** Higher accuracy for selected quantiles
- **Privacy:** a differentially private version of SplineSketch that preserves high accuracy